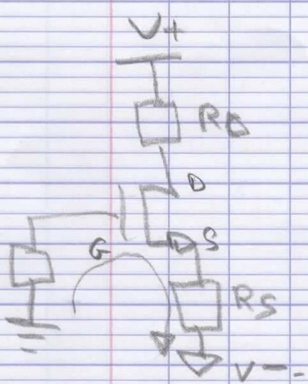
Rappel :
$V_{tn} = 1{,}3\,V$
$k_n = 2\,mA/V^2$
$\lambda = 0{,}02\,V^{-1}$

$$Z_C = \frac{1}{jC\omega} \Rightarrow \begin{cases}\omega = 0 & Z_C = \infty \\ \omega = \infty & Z_C = 0\end{cases}$$

$$Z_L = jL\omega \Rightarrow \begin{cases}\omega = 0 & Z_L = 0 \\ \omega = \infty & Z_L = \infty\end{cases}$$

Q 1) Schéma en DC :



$I_G = 0\,A$
$R_G = 0$

Q2)
$$\begin{cases} I_d = \frac{1}{2} k_n \underbrace{(V_{GS} - V_{tn})}_{V_{OV}}^2 \\ g_m = \frac{\partial i_D}{\partial V_{GS}} \end{cases}$$

$$g_m = \frac{\partial \frac{1}{2} k_n (V_{GS} - V_{tn})^2}{\partial V_{GS}}$$

$$g_m = \frac{1}{2} k_n \cdot 2 (V_{GS} - V_{tn}) = k_n (V_{GS} - V_{tn})$$

Mais $k_n = \frac{2 I_d}{(V_{GS} - V_{tn})^2}$

Soit $g_m = \frac{2 I_d}{(V_{GS} - V_{tn})}$

$$V_{GS} - V_{tn} = \frac{2 I_d}{g_m} \Rightarrow V_{GS} = \frac{2 I_d}{g_m} + V_{tn}$$

Q3 : $r_o = \frac{1}{\lambda I_d}$ ou $r_o = \frac{V_A}{I_d}$

$$V_{OV} = \sqrt{\frac{2 I_d}{k_n}} = 1\,V$$

$$g_m = \frac{2 \cdot 1 \cdot 10^{-3}}{1} = 2\,mA/V$$

$$r_o = \frac{1}{0{,}02 \cdot 1 \times 10^{-3}} = 50\,k\Omega$$

$V_{OV} = V_{GS} - V_{tn}$

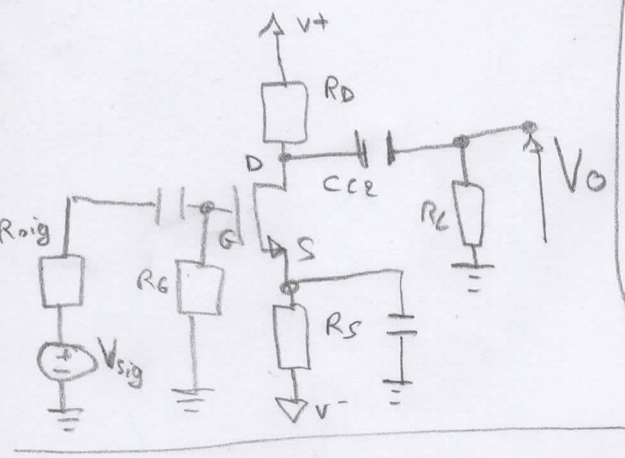
$V_{GS} = V_{OV} + V_{tn} = 1 + 1{,}3 = 2{,}3$

Calcul de RS :

$$V_G - V_{GS} - V_{RS} = V^-$$
$$-V_{RS} = V^- - V_G + V_{GS}$$
$$V_{RS} = -V^- + V_G - V_{GS}$$
$$R_S = \frac{-V^- + V_G - V_{GS}}{I_D}$$
$$R_S = \frac{15 + 0 - 2{,}3}{1 \cdot 10^{-3}} = 12{,}7\,k\Omega$$

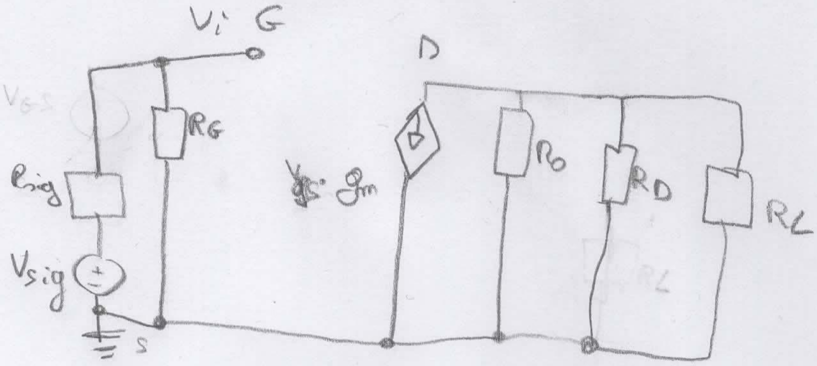
Signal AC

Modèle de Base



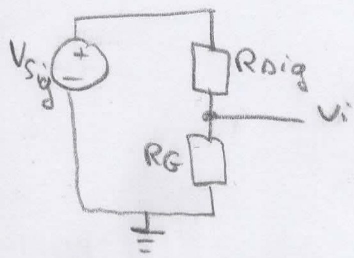
Petit signal

Q1



$$Av = \frac{v_o}{v_i} = -g_m\,(R_D \parallel R_L \parallel R_o)$$

Q2)



$$V_i = V_{sig} \cdot \frac{R_G}{R_G + R_{sig}}$$

$$\frac{V_i}{V_{sig}} = \frac{R_G}{R_G + R_{sig}} = \frac{10k}{10k + 15k} = \quad 0{,}995$$

Q3) $R_{eq} = R_D \parallel R_L \parallel r_o = \frac{1}{R_D} + \frac{1}{R_L} + \frac{1}{r_o}$

$Av = -g_m \cdot R_{eq}$

$R_{eq} = \frac{A_m}{-g_m} = \frac{-5}{2 \cdot 10^{-3}} = -2{,}5\ k\Omega$

Donc $\frac{1}{R_D} = \frac{-1}{R_L} - \frac{1}{R_O} + \frac{1}{R_{eq}}$

$\frac{1}{R_D} = 0{,}28\ \Omega^{-1}$

$R_D = 3{,}57\ k\Omega$

Q4) $V_D = V^+ - R_D \cdot I_d$

$= 15 - 3{,}57 \cdot 10^{+3} \cdot 1 \cdot 10^{-3}$

$= 11{,}43\,V$

Condition :

$V_{DS} \geq V_{GS} - V_{tH}$

$V_D \geq V_G - V_{tH}$

$\downarrow$ 0V $\quad \downarrow$ 1,3 V

Condition vérifiée